need to traverse S_1's branch as well, because Z lies within it. As a matter of fact, the search region intersects another leaf, S_{10}, so we need to traverse a path to it: as you can see, point J is close to be Z's nearest neighbor, so it's not unlikely that we would find the true NN only in a later call.

We can easily extend the nearest neighbor search algorithm to return the n-th nearest neighbor: similarly to what we have done in chapter 9 for k-d trees, we just need to use a bounded priority queue that keeps at most n elements, and use the furthest of those n points as reference, computing the pruning distance as the distance from this point to the search target (as long as we have found less than n, the pruning distance will be infinity).

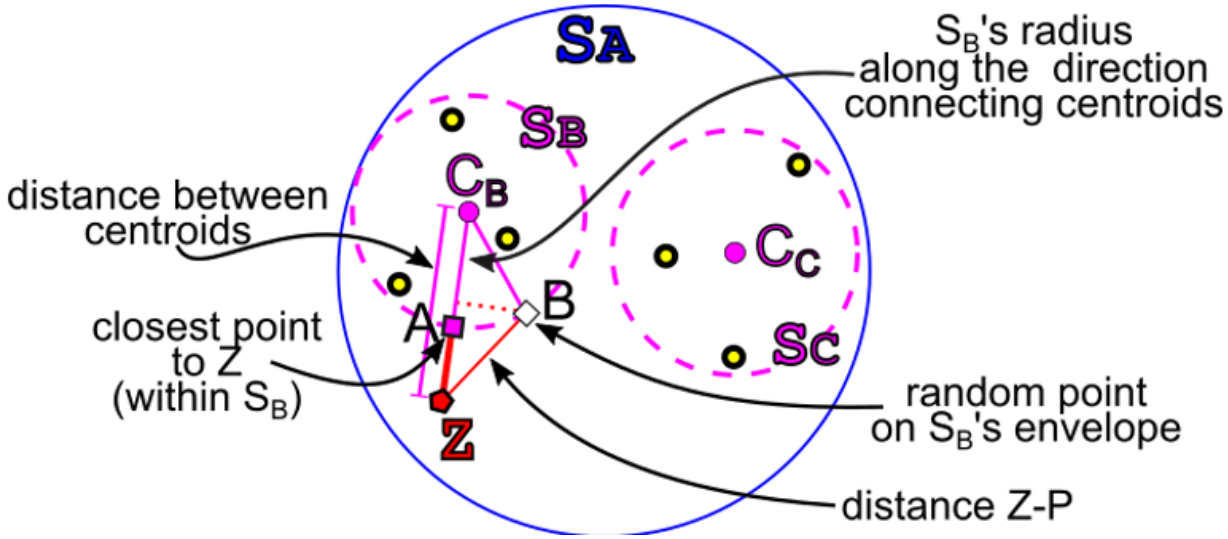


Figure 10.22 Minimum distance to a bounding envelope. Consider the triangle ZBC_B. Then, for the triangular inequality, $|C_BB|+|BZ|>|ZC_B|$, but $|ZC_B|==|ZA|+|AC_B|$ and $|AC_B|==|C_BB|$ (they are both